

Name: _____

FRACTIONS, DECIMALS AND PERCENTAGES (RECURRING DECIMALS)

Date: _____

LO: I can convert recurring decimals to fractions using algebraic methods.

What is a recurring decimal?

Recurring decimals have one or more digits that **REPEAT** forever. We write a dot above the repeating digit(s): $0.\dot{3}$ means $0.333\dots$ and $0.1\dot{2}$ means $0.121212\dots$

Key fact: Every recurring decimal can be written as an exact fraction. Terminating decimals have denominators whose only prime factors are 2 and 5.

Terminating vs recurring

●	$0.333\dots = \frac{1}{3}$	recurring (one digit)
●	$0.272727\dots = \frac{3}{11}$	recurring (two digits)
●	$0.1666\dots = \frac{1}{6}$	recurring (after 0.1)
●	0.25 is recurring	it terminates (= $\frac{1}{4}$)
●	$0.333\dots = \frac{33}{100}$	that equals 0.33 only

Key idea

Let x = the recurring decimal. Multiply by 10 (or 100, 1000...) to shift the repeating block. Subtract to eliminate the recurring part. Solve for x .

$$x = 0.2\dot{7} \quad 100x = 27.2\dot{7} \quad 99x = 27 \quad x = \frac{27}{99} = \frac{3}{11}$$

Multiply, subtract, simplify

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - one repeating digit

Convert $0.4\dot{4}$ to a fraction

$$\text{Let } x = 0.444\dots$$

$$10x = 4.444\dots$$

$$9x = 4, \text{ so } x = \frac{4}{9}$$

$$= \frac{4}{9}$$

One repeating digit: multiply by 10.
Subtract x from $10x$ to get $9x = 4$.

Example 2 - two repeating digits

Convert $0.6\dot{3}$ to a fraction

$$\text{Let } x = 0.636363\dots$$

$$100x = 63.636363\dots$$

$$99x = 63, \text{ so } x = \frac{63}{99}$$

$$= \frac{7}{11}$$

Two repeating digits: multiply by 100.
Subtract: $99x = 63$. Simplify: $\frac{63}{99} = \frac{7}{11}$.

Example 3 - non-repeating start

Convert $0.16\dot{6}$ to a fraction

$$\text{Let } x = 0.1666\dots$$

$$10x = 1.666\dots, 100x = 16.666\dots$$

$$90x = 15, \text{ so } x = \frac{15}{90} = \frac{1}{6}$$

$$= \frac{1}{6}$$

The 1 doesn't repeat, but the 6 does. Use $100x - 10x = 90x$ to eliminate the recurring part.

Example 4 - three repeating digits

Convert $0.12\dot{3}$ to a fraction

$$\text{Let } x = 0.123123\dots$$

$$1000x = 123.123123\dots$$

$$999x = 123, \text{ so } x = \frac{123}{999}$$

$$= \frac{41}{333}$$

Three repeating digits: multiply by 1000.
Subtract: $999x = 123$. Simplify: $\frac{123}{999} = \frac{41}{333}$.

YOUR TURN – PRACTICE (deliberate practice)

1. Identify the recurring part

Write the first 6 decimal places and underline the repeating block.

- a) $0.\underline{7} = 0.\underline{\hspace{1cm}}$
 b) $0.\underline{18} = 0.\underline{\hspace{1cm}}$
 c) $0.\underline{41} = 0.\underline{\hspace{1cm}}$
 d) $0.\underline{523} = 0.\underline{\hspace{1cm}}$

2. One repeating digit fraction

Convert each recurring decimal to a fraction. Simplify where possible.

- a) $0.\underline{2}$
 b) $0.\underline{5}$
 c) $0.\underline{8}$
 d) $0.\underline{6}$

3. Two repeating digits fraction

Convert each recurring decimal to a fraction. Use $\times 100$ and subtract.

- a) $0.\underline{27}$
 b) $0.\underline{45}$
 c) $0.\underline{09}$
 d) $0.\underline{72}$

4. Non-repeating start fraction

Convert each recurring decimal to a fraction. Careful: part of the decimal doesn't repeat.

- a) $0.1\underline{3}$
 b) $0.8\underline{3}$
 c) $0.2\underline{36}$
 d) $0.4\underline{16}$

5. Mixed recurring decimal challenges

Convert to fractions. Simplify fully.

- a) $0.\underline{9}$ b) $0.1\underline{42857}$ c) $0.0\underline{3}$ d) $0.5\underline{4}$ e) $0.8\underline{3}$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $0.\underline{9} = 0.999... < 1$ ☐

Correct answer: _____

Reason: _____

b) $0.1\underline{2}$ means multiply by 100 to shift the block ☐

Correct answer: _____

Reason: _____

c) $0.\underline{3} = \frac{3}{10}$ ☐

Correct answer: _____

Reason: _____

d) $0.1\underline{6}$ needs $100x - 10x$ to eliminate the recurring part ☐

Correct answer: _____

Reason: _____

e) All fractions give terminating decimals ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Prove algebraically that $0.\underline{9} = 1$.

Let $x = 0.999...$ and use the multiply-and-subtract method. Explain why this proves they are equal.

Expression: _____

b) A student says " $\frac{1}{7} = 0.142857$ " (with a dot only over the 7).

Explain why this is wrong. What is the correct dot notation for $\frac{1}{7}$? How many digits repeat?

Expression: _____

TOP TIPS

Read the question carefully. Show all your working step by step. Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

